Forecasting trained pilots from a tactical viewpoint

What the experiments establish so far | 2026-11-02

19.1 m

MOTION AT +5 s

Mean position error
Original transformer

63.5%

SELECTION AT +5 s

Exact-label accuracy
Improved ensemble

+7.58 pts

ACCURACY GAIN

Versus original transformer
Same validation origins

One observation stream. Two distinct prediction tasks.

These are predictions of future motion and selected labels, not ratings of pilot competence.

B01 supplies the observations. A01 is the forecast target.

The trained policies generate behaviour; a separate model learns to forecast it.

B01 / SCRIPTED TACTICAL

Own state + relative target state
Available observations through time t

observes

A01 / TRAINED PILOT

One of ten frozen policies
Future state and selection are targets

FORECAST INPUTS

Observation history and causal geometry
No pilot ID or private skill decision

SUPERVISED OUTPUTS

Position / velocity at +1, +2, +5 seconds
Catalogue selection at those same times

Keep complete scenarios together when splitting the data

3,000 episodes = 300 starting scenarios × 10 trained pilots.

210

TRAINING GROUPS

2,100 episodes
Model fitting and internal selection

45

VALIDATION GROUPS

450 episodes
Reported development comparison

45

RESERVED TEST GROUPS

450 episodes
No reported test evaluation

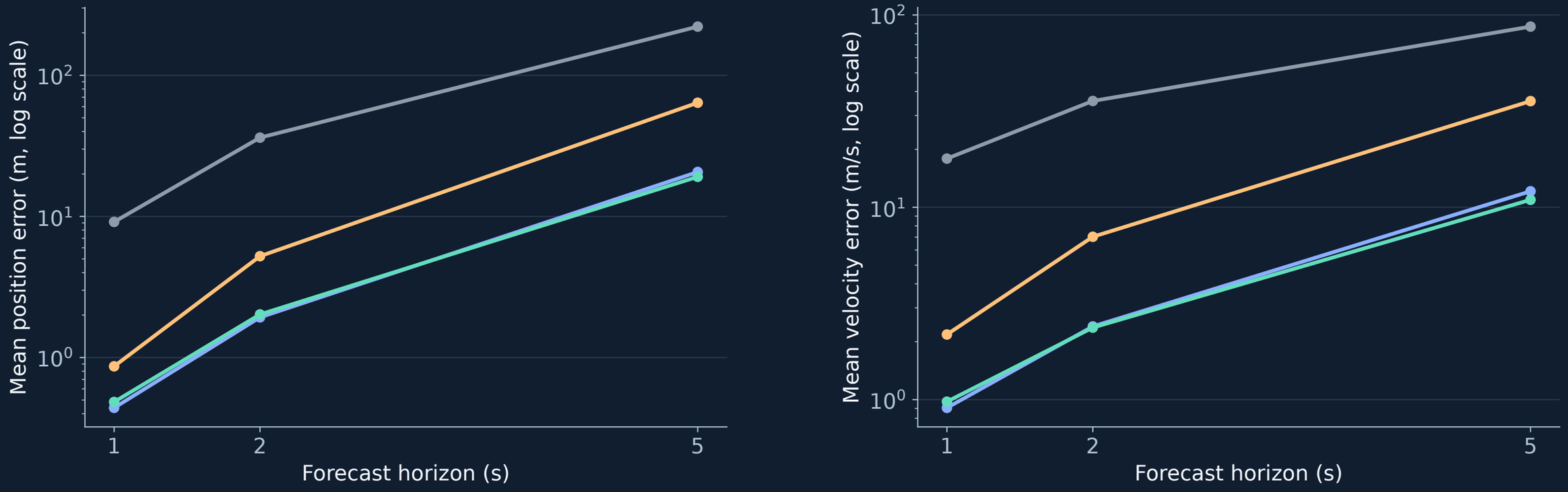
ONE GROUP CONTAINS ALL TEN PILOTS

Matching encounters stay in one partition. Overlapping windows are never randomly split.

10 Hz observations · 1 Hz selection labels · Exact +1 / +2 / +5 s targets

Learned motion forecasts improve on kinematic baselines

The GRU leads short-horizon position; the transformer leads at +5 seconds.



A skill forecast predicts a selected catalogue entry

It does not score competence, infer intent with certainty, or guarantee a distinct manoeuvre.

EXAMPLE ORIGIN: t = 6 s

Observe the available history up to 6 s.
Predict selections at 7, 8 and 11 s.

33 POSSIBLE LABELS

26 have truth support in validation.
Some names share executed guidance.

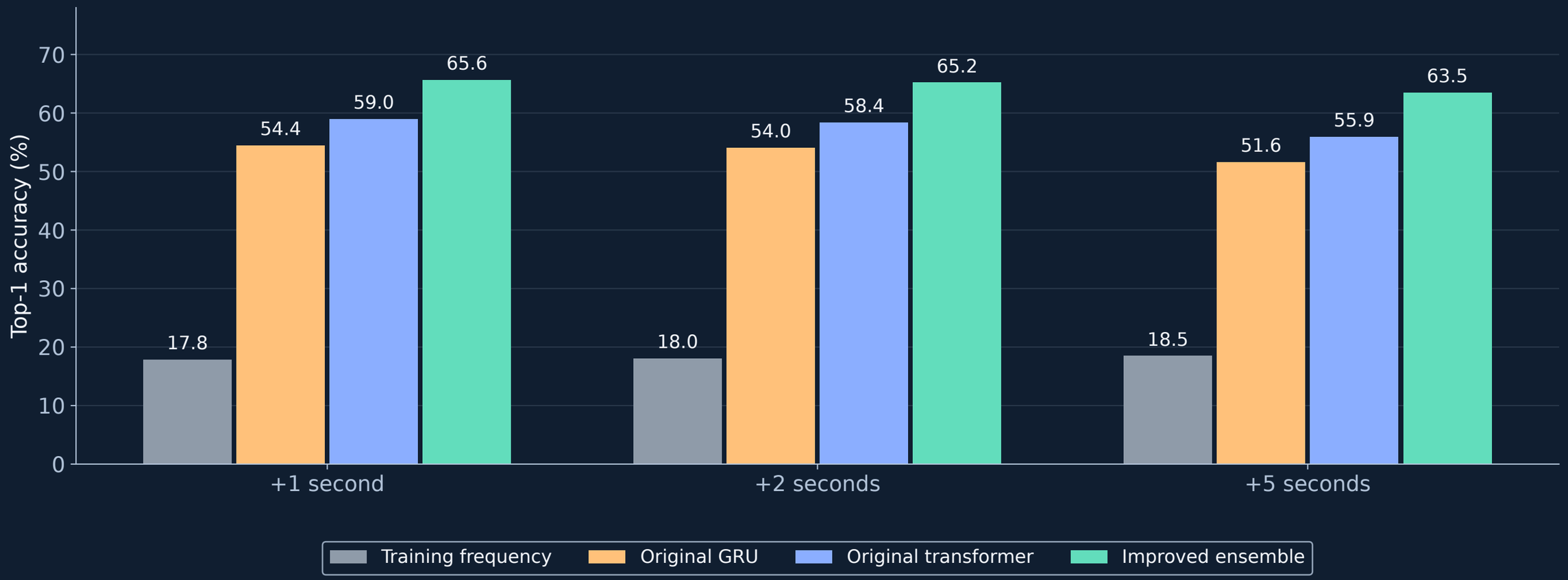
MOTION AND LABELS ANSWER DIFFERENT QUESTIONS

Motion: where will the trained aircraft be?
Catalogue: which entry will its policy select at that future instant?

Accurate motion does not imply that every private catalogue choice is identifiable.

Exact-label accuracy rises at all three forecast horizons

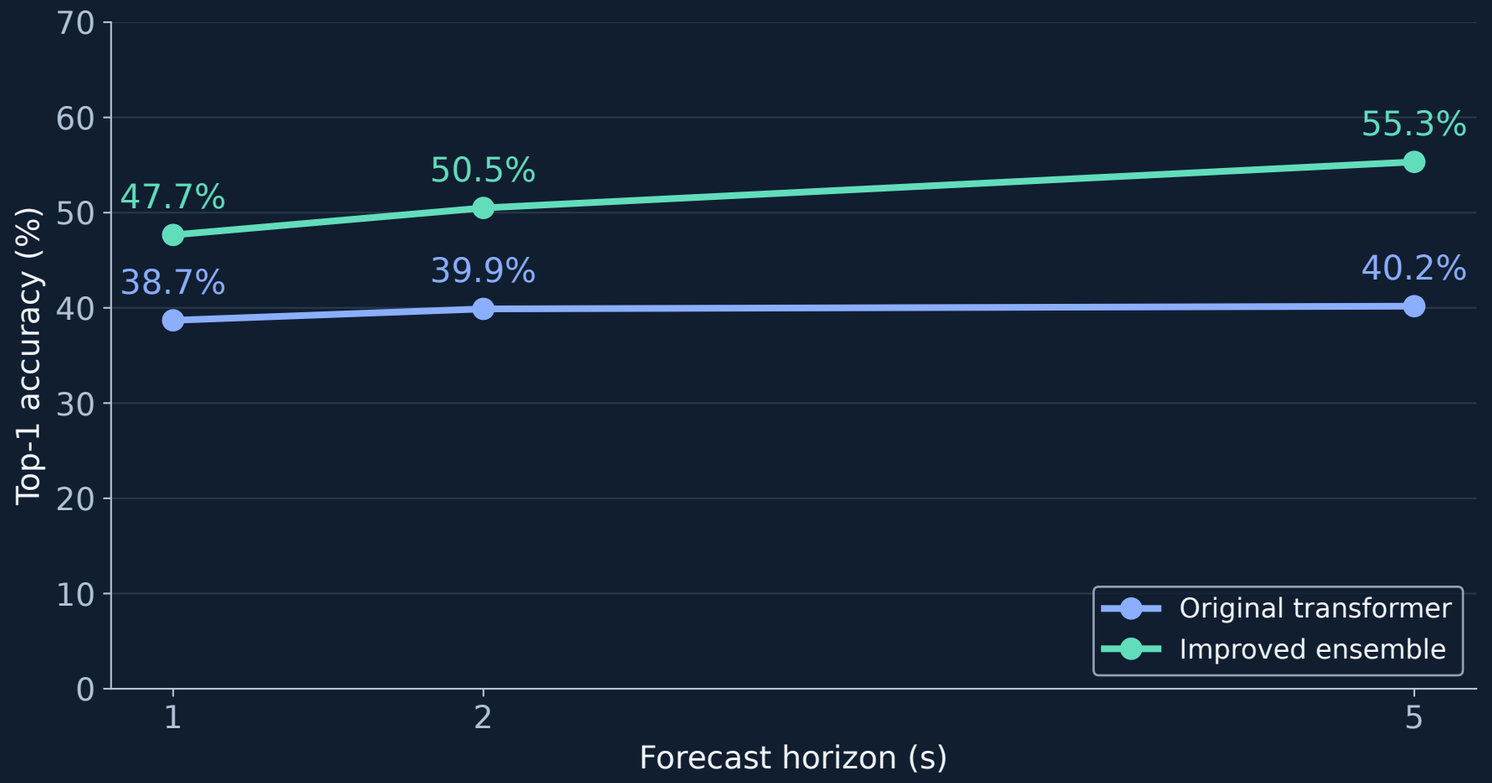
The new ensemble improves on both initial neural models and the frequency reference.



skill_*.csv | Same origins and horizon masks | Development validation; test reserved

The harder changed-selection subset improves too

Cases where the future selection differs from the true selection at the origin.



AT +5 SECONDS

40.2% → 55.3%
8,659 changed origins

INTERPRETATION

Useful gains beyond cases with the same endpoint label.

This is not a switch detector.

skill_transformer.csv + skill_ensemble.csv | Changed cohort | Development validation; test reserved

All ten pilots improve, with uneven gain sizes

Each cell is the ensemble's top-1 gain over the original transformer, in percentage points.

	+1 s	+2 s	+5 s
Pilot 001	+4.5	+6.1	+6.3
Pilot 002	+9.8	+10.2	+11.8
Pilot 003	+3.2	+3.2	+5.9
Pilot 004	+1.8	+1.8	+0.9
Pilot 005	+9.8	+10.5	+12.9
Pilot 006	+8.2	+8.0	+7.2
Pilot 007	+2.1	+0.8	+0.8
Pilot 008	+7.6	+8.4	+7.3
Pilot 009	+7.6	+7.6	+10.0
Pilot 010	+11.6	+11.1	+12.4

BROAD DIRECTION OF IMPROVEMENT

30 of 30 pilot / horizon cells improve.
The gains are not uniform.

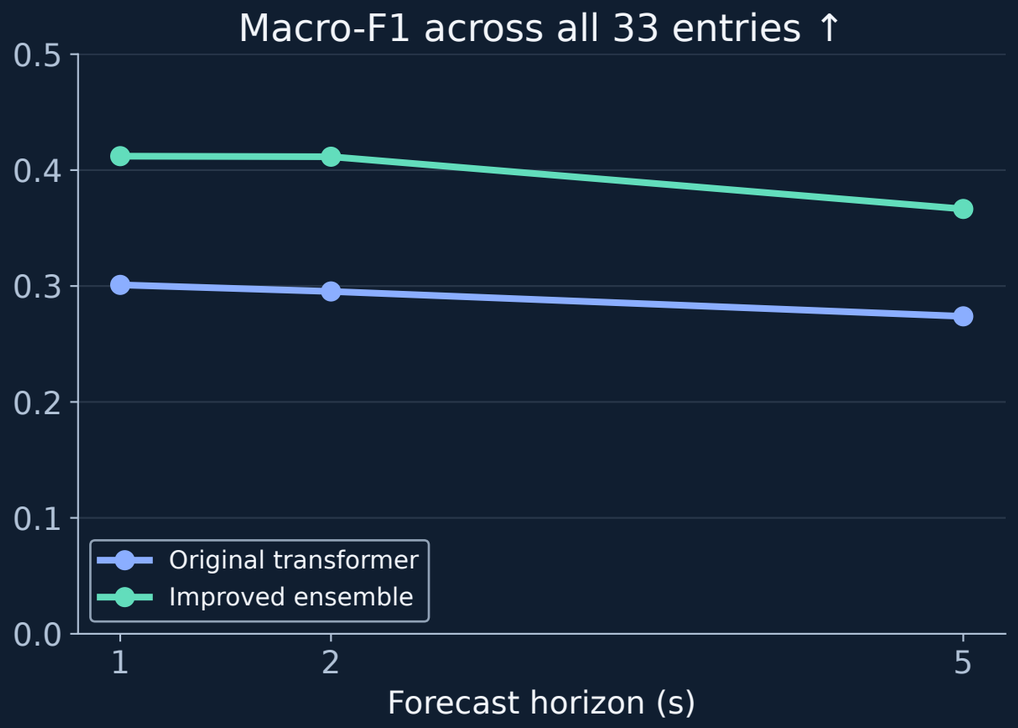
SCOPE OF THE CLAIM

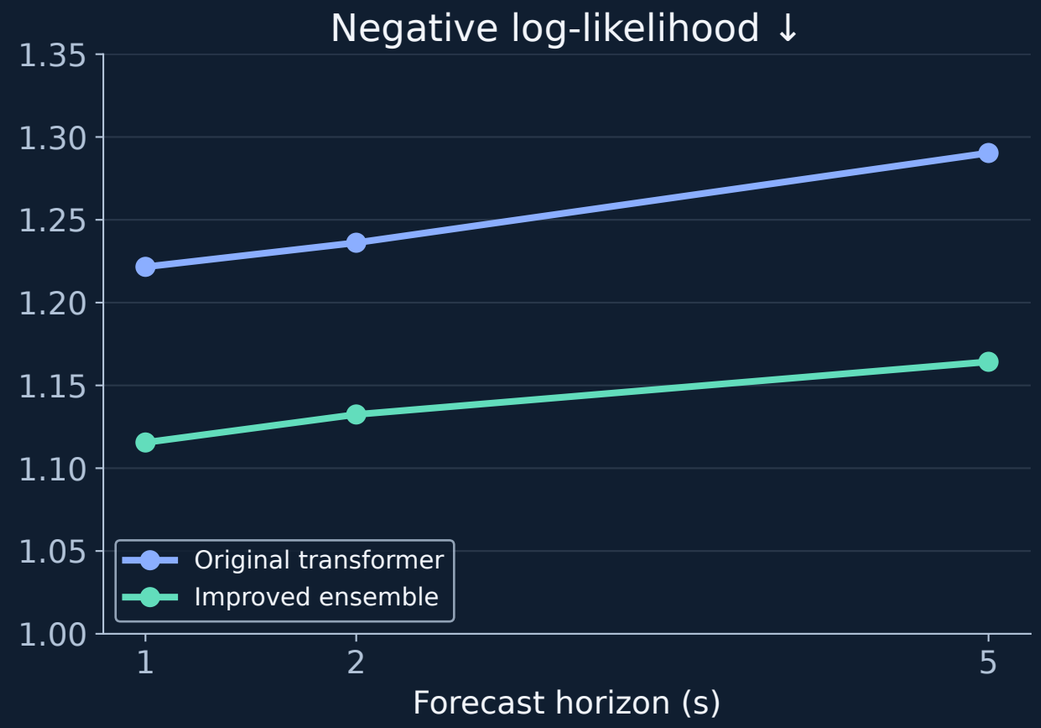
New scenarios for these ten pilots.
No unseen-pilot evaluation.
Pilot identity is not a model input.

skill_transformer.csv + skill_ensemble.csv | Per-pilot all cohort | Development validation; test reserved

The gain also appears in class balance and log-loss

Overall top-3 accuracy improves; improvements are not universal across every subset.





Improved top-3 accuracy: 90.1% at +1 s · 89.8% at +2 s · 89.2% at +5 s

The selected improvement combines four model variants

Geometry, longer history, regularization and uniform probability averaging were explored together.

91 OBSERVABLE / DERIVED FEATURES

Relative geometry, speed and turn cues
Causal acceleration; no future trajectory

20 s AND 60 s HISTORIES

Three 20-second temporal MLP variants
One 60-second temporal MLP variant

3 FIXED SEEDS PER VARIANT

12 models; equal probability weights
Stop-group temperature calibration

THE COST OF THE IMPROVEMENT

29 million parameters; ~116 MB
Local CPU check: ~20 ms per forecast

All new model selection stayed inside the training groups

Two internal partitions informed the final configuration before the validation comparison.

147

FIT GROUPS

Gradients + normalization

21

STOP GROUPS

Epoch choice + calibration

42

SELECT GROUPS

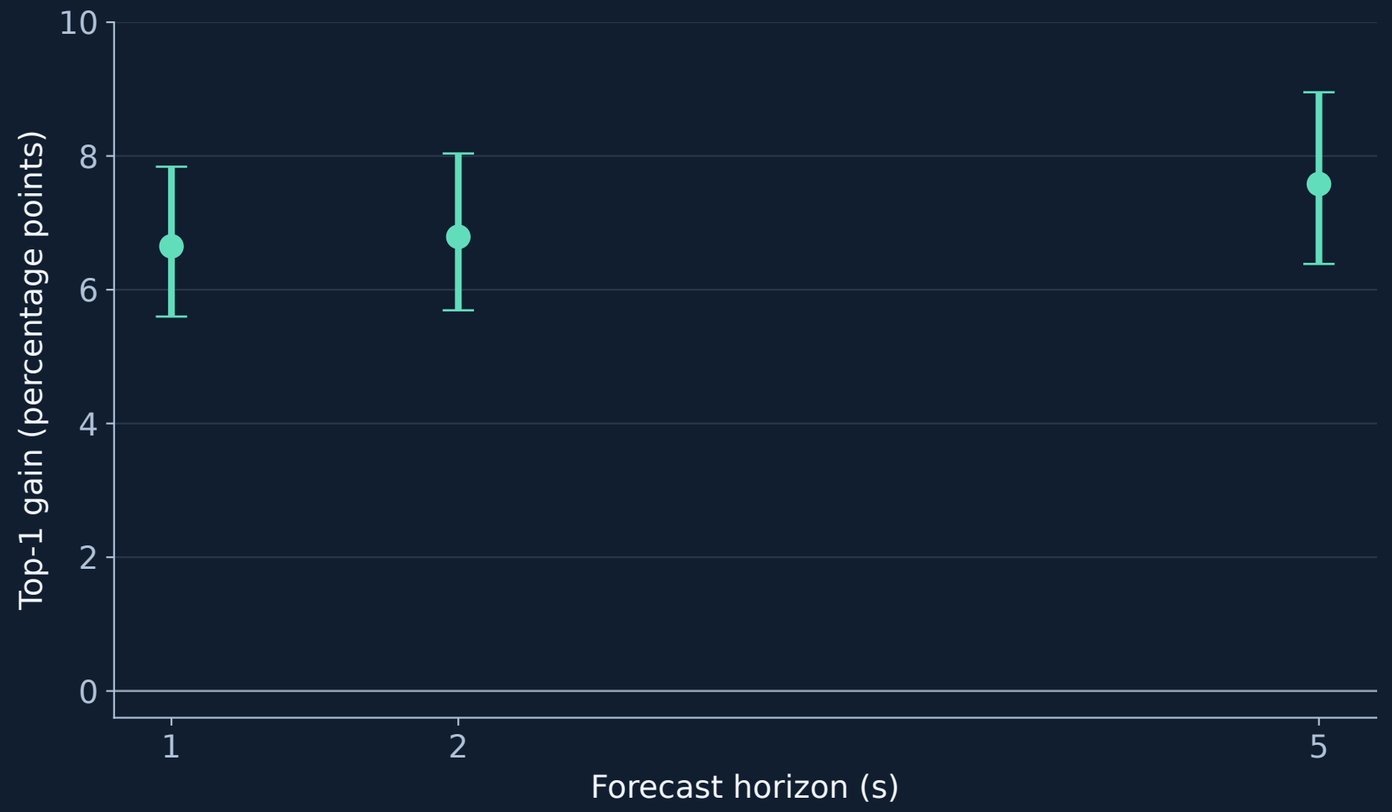
Candidate comparison

THEN FREEZE → REFIT → COMPARE

Fix components, epochs, temperatures and seeds; refit on all 210 training groups.
Evaluate the frozen ensemble on the original 45 validation groups.

Accuracy gains remain positive within the 95% intervals

95% bootstrap intervals preserve all pilots and overlapping windows within each scenario.



GAINS AND 95% INTERVALS

+1 s: +6.65 points [5.60, 7.84]
+2 s: +6.79 points [5.69, 8.04]
+5 s: +7.58 points [6.39, 8.95]

DEVELOPMENT EVIDENCE

No correction for all model selection.
Not an untouched-test claim.
Not a guarantee for new pilots.

scenario_intervals.csv | 45 groups; 5,000 paired bootstrap draws | Development validation; test reserved

Useful forecasts; clear limits on what is established

Motion and exact-label forecasting both show learnable structure in these recorded encounters.

SUPPORTED BY THE RESULTS

Neural motion forecasts beat kinematic baselines.
The skill ensemble gains ~7 accuracy points.
Gains reach every known pilot and changed cases.
Original results remain preserved.

STILL TO ESTABLISH

Performance on the reserved test scenarios.
Generalisation to unseen trained policies.
Robustness to noisy or missing observations.
Whether fewer models retain the gain.

Next: predeclare the final test comparison, then evaluate once.

The companion notes and portable evidence keep each claim tied to its metric and scope.